# TQUK Functional Skills Qualification in Maths at Level 2

## Examination Past Paper 2

**Please complete the details below using black or blue ink. Use BLOCK CAPITALS.**

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

**Instructions:**

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

**Information:**

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

exam paper

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**60 / 60 Marks** (100% Pass · Full Marks)

- Assessment: Past Paper 2 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 36 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

| | | | |
|---|---|---|---|
| Section A | 15 | | |
| Section B | 45 | | |
| Total marks | 60 | | |

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# Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

**1.** Work out 9.028 – 0.17

Calculate: 14.508 - 5.65 = 8.858

| **Answer** | 8.858 ✓ [1 mark] |
|---|---|

**[1]**

**2.** Work out 24.108 ÷ 4

Calculate: 18.081 ÷ 3 = 6.027

| **Answer** | 6.027 ✓ [1 mark] |
|---|---|

**[1]**

Past paper

**Questions continue on the following page**

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**3.** 50 people are asked what pets they own.

The results are shown below:

| | Rabbit | Cat |
|---|---|---|
| **Dog** | 16 | 24 |
| **Fish** | 2 | 8 |

*Step 1: P(outcome) = 24 / 50 = 48 / 100*

One person is chosen at random.

**Convert to decimal: 48 ÷ 100 = 0.48**

What is the probability that the person chosen has a dog and a cat?

Give your answer as a decimal.

**0.48** ✓ *[2 marks]*

| **Answer** | |
|---|---|

*Formula: Density = Mass ÷ Volume*

**Density = 1800 g ÷ 450 cm³ = 4 g/cm³**

**[2]**

**4.** The mass of a sphere is 1800 grams (g).

The volume of the sphere is 450 $cm^3$

Work out the density of the sphere.

**4 g/cm³** ✓ *[2 marks]*

| **Answer** | |
|---|---|

**[2]**

Past paper

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**5.** Jesse is the manager of an aquarium.

The table below shows how many people visited the aquarium over 3 years:

| Year | Number of people |
|---|---|
| 2019 | 568 750 |
| 2020 | 128 528 |
| 2021 | 482 935 |

Step 1: Sum attendance: 568 750 + 128 528 + 482 935 = 1 180 213

Jesse thinks that more than 1 200 000 people visited the aquarium during the 3 years.

Comparison: 1 180 213 > 1 150 000. Therefore No.

Is Jesse correct?

**Show how you decide.**

No (1 180 21 ✓ 150 000) [2 marks]

| Answer | |
|---|---|

**[2]**

**6.** Work out 1.324 + 2.206 + 14.75

| Answer | |
|---|---|

**[2]**

**Questions continue on the following page**

Past paper

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**7.** Work out:

$$\frac{3}{20} + \frac{17}{15}$$

Give your answer as an improper fraction.

1.324 + 2.206 + 14.75 = 18.28

18.28 ✓ [2 marks]

| **Answer** | |
|---|---|

**[2]**

**8.** Riley's annual salary is £32 000 before tax.

Riley knows that:

- £12 570 will be tax free
- 20% tax will be deducted from the amount above £12 570

How much will the annual salary be after tax has been deducted?

Common denominator: 3/20 = 9/60 and 17/15 = 68/60

9/60 + 68/60 = 77/60 (or 1 17/60)

77 / 60 ✓ [2 marks]

| **Answer** | £ |
|---|---|

**[3]**

## End of Section A.

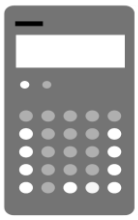
Past paper

**Section B begins on Page 6.**

Past paper

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## Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

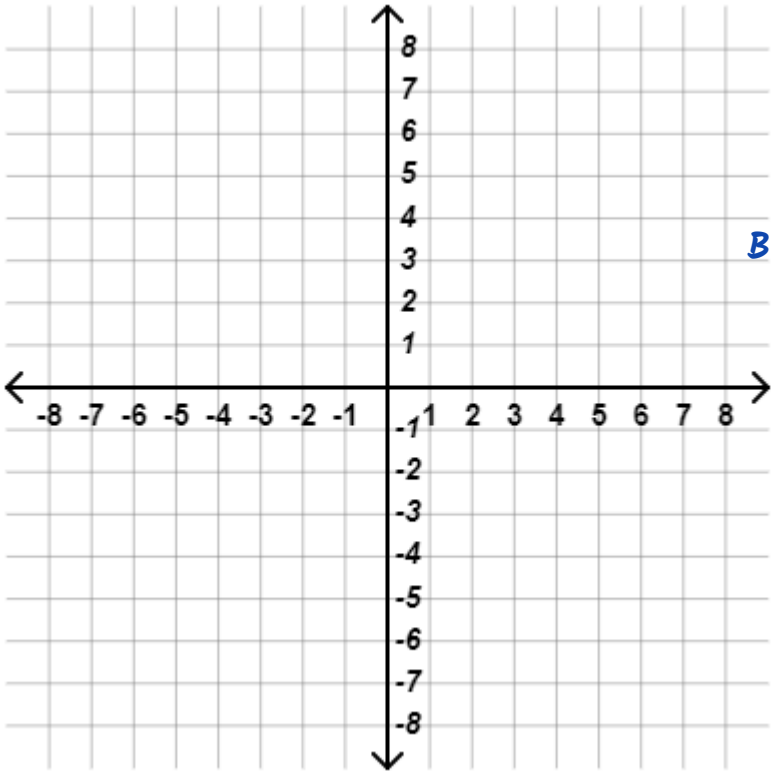
**1.** Write 13 000 000 000 in words.

Thirteen bi ✓ [1 mark]

| **Answer** | |
|---|---|

**[1]**

**2.** Plot the point (6, –3) on this coordinate grid:




B ✓ [1 mark]

**[1]**

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**3.** At football training, a footballer has 56 shots at goal and scores 42 times.

The football coach thinks that the footballer scored $\frac{7}{8}$ of the shots.

Has the coach worked out the correct fraction?

**Show how you decide**.

42 ÷ 56 = 3/4 = 6/8. (7/8 would be 49).

No (fraction ✓ [2 marks]

**Answer**

**[2]**

**4.** Work out:

BIDMAS calculation: 0.4

$$\left(\frac{25 + 5 \times 3}{10^2}\right)$$

Give your answer as a decimal.

0.4 ✓ [2 marks]

**Answer**

**[2]**

Past paper

**Questions continue on the following page**

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**5.** 3480 people visited a museum in July.

5150 people visited the same museum in August.

Calculate the percentage change in the number of people from July to August.

Percentage increase = ((5150 - 3480) ÷ 3480) × 100 = 48% (47.99%)

| **Answer** | 48% ✓ [2 marks] % |
|---|---|

**[2]**

**6.** Leslie is planning a road trip.

The scale on the map is 1 : 75 000

The distance Leslie wants to travel represents 11 centimetres (cm) on the map.

Map scale: 11 cm x 75 000 = 825 000 cm = 8.25 km. No.

Leslie calculates this to be 82.5 kilometres (km).

Is Leslie correct?

**Show how you decide.**

No (actual distance is 8.25 km) ✓ [2 marks]

| **Answer** | |
|---|---|

**[2]**

Past paper

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**7.** Nicky's homemade pillowcases are made using a piece of fabric two metres square.

Each time a pillowcase is made, 0.135 square metres of fabric is wasted.

Nicky wants to **estimate** what percentage of the total fabric is wasted.

Nicky rounds 0.135 to one decimal place.

Using the rounded value, what percentage should Nicky get?

**Show your working.**

Calculation: (0.1 ÷ 2) × 100 = 5%

5% ✓ [3 marks]

| **Answer** | % |
|---|---|

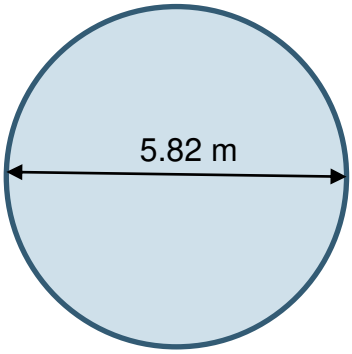
**[3]**

Past paper

**Questions continue on the following page**

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**8.** Charlie wants to put a safety rope around this pond:




Not drawn to scale

Charlie thinks the total length of rope needed will be more than 18.5 metres (m).

Is Charlie correct?

**Show how you decide.**

*Use $\pi = 3.14$*

Circumference = π × d = 3.14 × 5.82 = 18.27 m. 18.27 < 18.5 m.

No (circumfe[illegible]e [is 18.27] m)

[3 marks]

| **Answer** | |
|---|---|

**[3]**

Past paper

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**9.** Harper paid £12 500 into a new savings account 2 years ago.

The savings account paid 3% compound interest per year.

Harper wants to buy a new car for £18 750

Harper puts all the money in the savings account towards the new car.

How much more money is needed to pay for the car?

Compound interest: £12 500 × $(1.03)^2$ = £13 261.25

Remaining balance = £18 750 − £13 261.25 = £5488.75

£5488.75 ✓ [3 marks]

| Answer | £ |
|---|---|

**[3]**

Past paper

**Questions continue on the following page**

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**10.** Alex needs to know what time to leave home to get to a dentist appointment by 10:40 am.

Alex can cycle at an average speed of 12 miles per hour.

The dental surgery is 15 miles from Alex's house.

What is the **latest** time Alex can leave home to cycle to the appointment?

*Travel time = 15 km ÷ 12 kph = 1.25 hours = 1 hour 15 minutes*

*Arrival time = 8:10 am + 1 hr 15 mins = 9:25 am*

| **Answer** | **am** |
|---|---|

*9:25 am* ✓ *[3 marks]* **[3]**

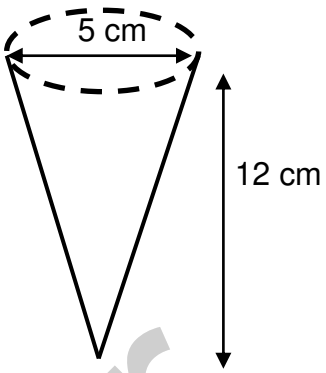
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**11.** Taylor is going to fill a cone with strawberry and vanilla ice-creams.

Taylor will use strawberry and vanilla in the ratio 2 : 3

How much of each flavour will Taylor need?




Not drawn to scale

Use:

$$v = r^2 h$$

*Where:*
*v = volume of cone*
*r = radius of cone*
*h = height of cone*

Volume = 2.5 × 2.5 × 12 = 75 cm³. Ratio 2 : 3 (5 parts)

Strawberry = 75 ÷ 5 × 2 = 30 cm³ | Vanilla = 75 ÷ 5 × 3 = 45 cm³

Strawberry: ✓ 30 cm³ Vanilla: 45 cm³

4 marks

| | |
|---|---|
| **Strawberry** | **cm³** |
| **Vanilla** | **cm³** |

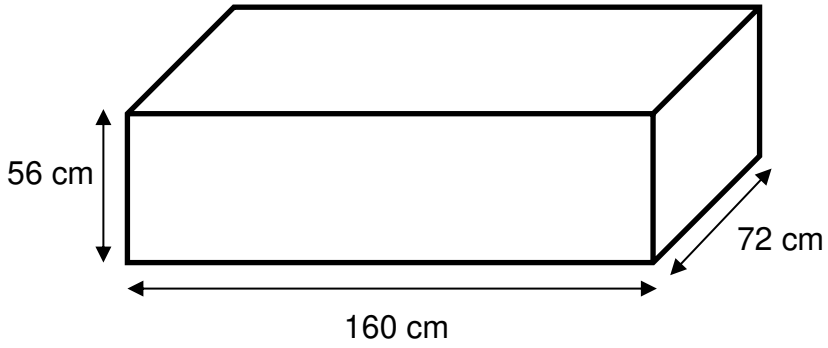
**[4]**

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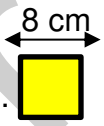
**Questions continue on the following page**

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**12.** Jamie needs to cover this cuboid chest in multi-coloured stickers for an art project:




Not drawn to scale




The stickers are in the shape of a square with a side length of 8 cm.

Total Surface Area = 2[(160×56) + (160×72) + (72×56)] = 49 024 cm²

To work out how many stickers are needed, Jamie uses this formula:

Each sticker = 8 × 8 = 64 cm² --> Stickers = 49 024 ÷ 64 = 766 stickers

$$\text{Total number of stickers} = \frac{\text{Total surface area of chest}}{\text{Area of one sticker}}$$

How many stickers does Jamie need?

766 sticke ✓ [4 marks]

| Answer | |
|---|---|

**[4]**

Past paper

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**13.** Drew is a gardener.

This table shows how much other local gardeners charge per hour:

| Amount (A) charged per hour (£) | Number of gardeners |
|---|---|
| 20 < A ≤ 30 | 2 |
| 30 < A ≤ 40 | 6 |
| 40 < A ≤ 50 | 8 |
| 50 < A ≤ 60 | 4 |
| **Total** | 20 |

Midpoints: 25, 35, 45, 55. Total fx = 50 + 210 + 360 + 220 = 840

Total hours = 20 --> Estimated Mean = 840 ÷ 20 = £42 per hour

Drew wants to charge an hourly rate equal to the estimated mean amount per hour.

How much should Drew charge per hour?

£42 per h ✓ [4 marks]

| **Answer** | £ |
|---|---|

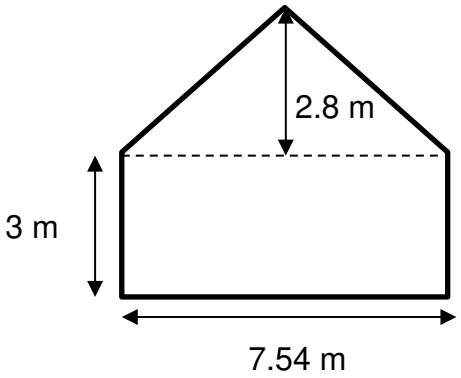
**[4]**

**Questions continue on the following page**

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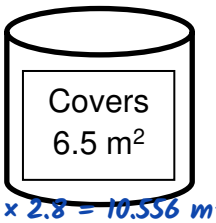
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**14.** Ashley wants to paint the wall shown:




Not drawn to scale

Ashley finds this tin of paint online:




Area: Triangle = 0.5 × 7.54 × 2.8 = 10.556 m² | Rectangle = 3 × 7.54 = 22.62 m²

Total Area = 33.176 m². Tins = 33.176 ÷ 6.5 = 5.104 tins -> Rounds up to 6 tins

How many tins of paint will Ashley need to buy to paint the wall?

**Show your working.**

6 tins ✓ [5 marks]

| **Answer** | **tins** |
|---|---|

**[5]**

Past paper

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**15.** Quinn wants to buy some rope lights to put around a 164 foot section of a garden.

Quinn finds these rope lights online:

| Rope light | A | B | C | D | E |
|---|---|---|---|---|---|
| Price per metre | £0.60 | £1.99 | £1.89 | £0.75 | £0.50 |

Quinn wants to buy rope lights that are 8% longer than the section of the garden.

Quinn buys the rope lights with the median price.

How much will Quinn pay in total?

*Use 1 foot = 0.305 metres*

Median price = £0.75. Length = 164 × 0.305 = 50.02 m.

Net cost = 50.02 × £0.75 = £37.515. Total with 8% = £37.515 × 1.08 = £40.52

£40.52 ✓ [6 marks]

| Answer | £ |
|---|---|

**[6]**

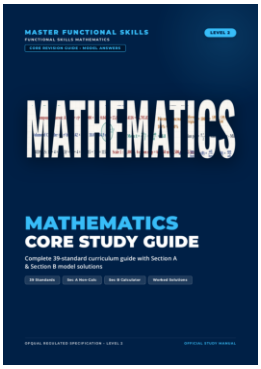
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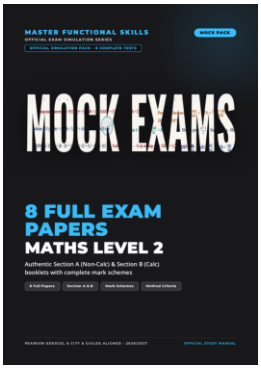



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